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Probability Distribution Function and types of distribution

⇒ Prob. distⁿ : PDFⁿ describes how the probabilities are distributed over the values of a random variable.

⇒ PDFⁿ are of 2 types :

① Prob. Mass Fuⁿ (PMF) → used for discrete random var.

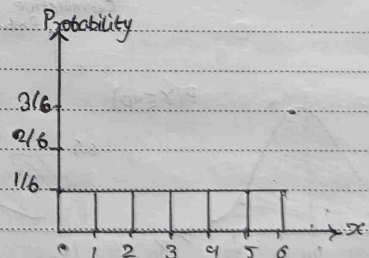
② " Density Fuⁿ (PDF) → " " continuous " "

⇒ PMF :

Rolling a die : $\{1, 2, 3, 4, 5, 6\}$

$$\therefore P(1) = \dots = P(6) = 1/6$$

∴ PMF :



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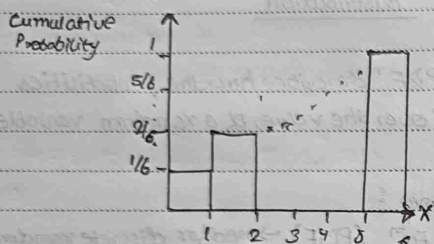
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* Prob. Density $\rightarrow$ gradient of CDF

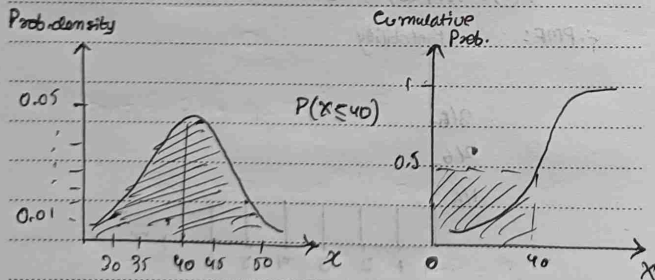
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$\Rightarrow$ Cumulative Density Funⁿ (CDF):



$$\therefore P(X \leq 2) = P(X=1) + P(X=2) \\ = 1/6 + 1/6 = 1/3$$

$\Rightarrow$ Prob. Density Funⁿ: distⁿ of cont. random variable
 $X = \text{Ages} = \{ \dots \}$



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$\Rightarrow$ PDF properties:

- ① always non-negative, $f(x) \geq 0 \forall x$
- ② the total area under PDF curve is equal to 1

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Types of Prob. distⁿ:

- ① Bernoulli distribution $\rightarrow$ outcomes are binary (PMF)
- ② Binomial " $\rightarrow$ PMF
- ③ Normal/Gaussian $\rightarrow$ PDF
- ④ Poisson distribution $\rightarrow$ PMF
- ⑤ Log Normal " $\rightarrow$ PDF
- ⑥ Uniform " $\rightarrow$ PMF

Bernoulli Distribution:

$\rightarrow$ thus for discrete random variable

$\Rightarrow$ It's the simplest discrete prob. distⁿ (hence uses PMF).

The outcomes are binary.

e.g. Tossing a coin

{H, T}

$$\therefore P(X=H) = 0.5 = P$$

$$P(X=T) = 1 - 0.5 = 0.5 = q = 1 - P$$

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⑤ Person with pass or fail

$$P(X=Fail) = 0.4$$

$$P(X=Pass) = 1 - 0.4 = 0.6$$

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⇒ Parameters:

$$0 \leq P \leq 1$$

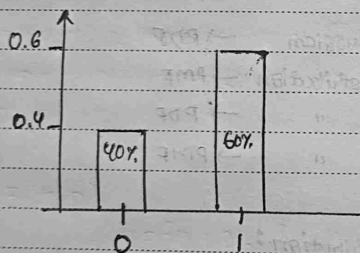
$$q = 1 - P$$

$$K = \{0, 1\} \Rightarrow 2 \text{ outcomes}$$

⇒ PMF: Company has launched a new smartphone 'A'.

$$\text{Use} = 60\% = P$$

$$\text{Not use} = 40\% = 1 - P$$



$$\text{PMF} = P^K * (1-P)^{1-K}$$

$$\text{if } K=1$$

$$\therefore P(K=1) = P^1 (1-P)^{1-1} = P$$

$$\text{if } K=0$$

$$\therefore P(K=0) = P^0 (1-P)^{1-0} = 1-P = q$$



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Simplified:

$$\begin{cases} q = 1 - P & \text{if } K=0 \\ P & \text{if } K=1 \end{cases}$$

⇒ Mean of Bernⁿ Distⁿ:

$$E(X) = \sum_{K=0}^1 K \cdot P(K), \quad K = \{0, 1\}$$

$$\text{if } K=0 \Rightarrow P(K=0) = 0.4$$

$$K=1 \Rightarrow P(K=1) = 0.6$$

$$\therefore E(X) = 0 \times 0.40 + 1 \times 0.60 = 0.60 = P$$

⇒ Median of Bernⁿ Distⁿ:

$$\text{Median} = \begin{cases} 0 & \text{if } P < 1/2 \text{ (or } q > P) \\ [0, 1] & \text{if } P = 1/2 \text{ (or } q = P) \\ 1 & \text{if } P > 1/2 \text{ (or } q < P) \end{cases}$$

⇒ Mode:

$$P > q \Rightarrow P \text{ is mode}$$

$$q > P \Rightarrow q \text{ is mode}$$



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⇒ Variance :

$$\sigma^2 = 0.40 \times (0-0.6)^2 + 0.60 \times (1-0.6)^2 = 0.24$$

$$\therefore \sigma^2 = pq \Rightarrow \sigma = \sqrt{pq}$$

Binomial Distribution :

- ⇒ (a) work with discrete random variable
- (b) every outcome of the experiment is binary
- (c) these expts are performed for n trials

⇒ Notations :

$$B(n, p)$$

$n \in \{0, 1, 2, \dots\} \Rightarrow$ no. of trials or expts.

$p \in [0, 1] \Rightarrow$ success prob. for each trial

$$q = 1 - p$$

Also, $K \in \{0, 1, 2, 3, \dots, n\} \Rightarrow$ No. of successes

$$\Rightarrow \text{PMF: } P_x(K, n, p) = {}^n C_K p^K (1-p)^{n-K}$$

where, $K = 0, 1, 2, \dots, n$

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$$\Rightarrow {}^n C_K = \frac{n!}{K!(n-K)!}$$

→ binomial coeff.

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⇒ Mean : $\pi \cdot p$

⇒ Variance : $\pi \cdot p \cdot q$

⇒ Std. dev : $\sqrt{\pi p q}$

e.g., coin flip

No. of trial (n) = 5

Pro. of success (p) = 0.5

No. of success (K) = 0 to 5

$$Pr(\text{exactly 3 heads in 5 flips}) \Rightarrow Pr(X=3) = {}^5 C_3 (0.5)^3 (0.5)^{5-3} = 0.3125$$

Poisson Distribution :

⇒ (a) Discrete random variable (PMF)

(b) describes the no. of events occurring in a fixed time intervals.

⇒ PMF :

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

⇒ Mean :

$$E(X) = \mu = \lambda \cdot t$$

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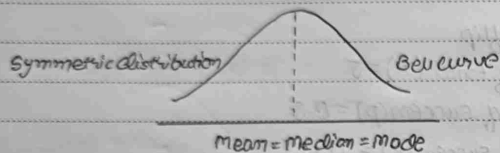
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↑ time interval
expected no. of events occur at
every time interval

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Normal Gaussian Distribution: e.g., heights of student in a class

⇒ (a) Continuous prob. distⁿ for cont. random variable



⇒ $N(\mu, \sigma^2)$

$\mu \in \mathbb{R}$ = mean

$\sigma^2 \in \mathbb{R} > 0$ = variance

$x \in \mathbb{R}$

$\sigma^2 \mu$
spread

⇒ PDF:

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2}$$

⇒ Mean:

$$\mu = \sum_{i=1}^n \frac{x_i}{n}$$

⇒ variance:

$$\sigma^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{n}$$



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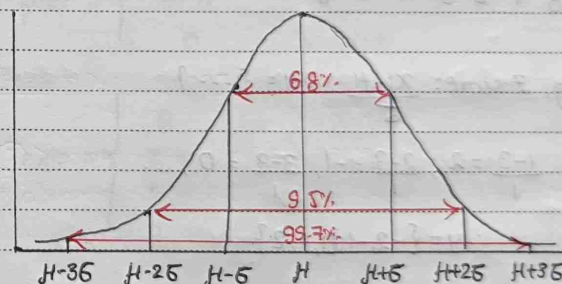
⇒ Empirical rule: 68-95-99.7 Rule

$X = \{ \dots \}$ ⇒ say it follows Normal gaussian distⁿ

$$P_x(\mu - \sigma \leq X \leq \mu + \sigma) \approx 68\%$$

$$P_x(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 95\%$$

$$P_x(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 99.7\%$$



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mean = 0
st. dev = 1

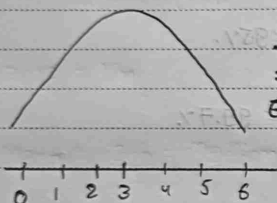
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Standard Normal Distribution:

$$\Rightarrow X = \{1, 2, 3, 4, 5\}$$

$$\therefore \mu = 3$$

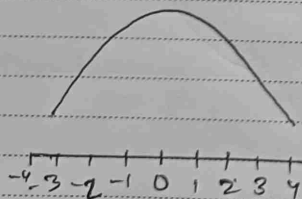
$$\sigma = 1.414 \approx 1$$



Using, Z-score: $\frac{X_i - \mu}{\sigma} \Rightarrow Y = \{\dots\}$

$$\therefore \frac{1-3}{1} = -2, \frac{2-3}{1} = -1, \frac{3-3}{1} = 0, \dots$$

$$\therefore Y = \{-2, -1, 0, 1, 2\}$$



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Uniform Distribution:

① Cont. uniform distⁿ (pdf)

② Discrete uniform distⁿ (pmf)

→ Continuous uniform distⁿ → for cont. random variable

$U(a, b)$

where, $a, b \Rightarrow -\infty < a < b < \infty$

$$\therefore \text{pdf} = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

$$\text{cdf} = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$$

$$\text{Mean} = \frac{1}{2}(a+b) = \text{Median}$$

$$\text{variance} = \frac{1}{12}(b-a)^2$$

$$P_x = \frac{(x_2 - x_1) \times L}{b-a}$$

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$\Rightarrow$ discrete uniform distⁿ $\rightarrow$ discrete random var.
 $\rightarrow$ pmf

$$1/n = b - a + 1$$

$$u(a, b)$$

where, $b > a$

$$\text{pmf} = 1/n$$

$$\text{mean} = \frac{a+b}{2} = \text{Median}$$

Log Normal Distribution $\rightarrow$ it's right skewed distⁿ

$\Rightarrow X \approx \log \text{Normal dist. } (\mu, \sigma)$

$\downarrow$

$$Y \approx \log(x) \text{ (or } \ln(x))$$

Normal distⁿ

Also,

$$X \approx \exp(Y) \Rightarrow \log \text{ normal distributed}$$

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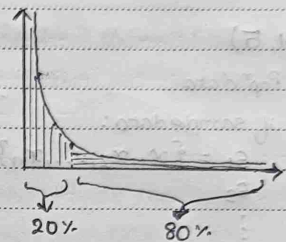
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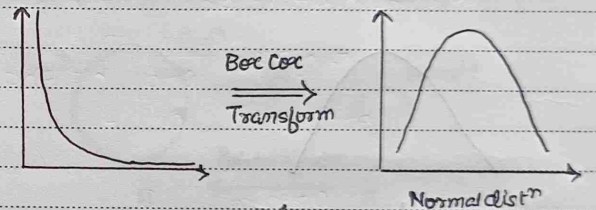
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Power Law Distribution $\rightarrow$ 80-20% Rule

$\Rightarrow$



$\Rightarrow$



Pareto Distribution

$\Rightarrow$ Non gaussian distribution.

$\Rightarrow$ It follows power law distⁿ

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Central Limit Theorem:

$\Rightarrow$ If $X \sim N(\mu, \sigma)$

Popⁿ data

If sampled data:

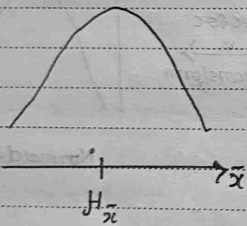
$$S_1 = \{x_1, x_2, \dots, x_n\} = \bar{x}_1$$

S_2

$\vdots$

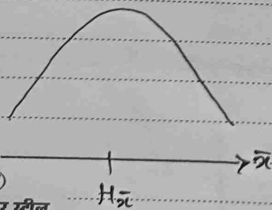
S_m

$\bar{x}_m$



$\Rightarrow$ If $X \not\sim N(\mu, \sigma)$

and $n > 30$



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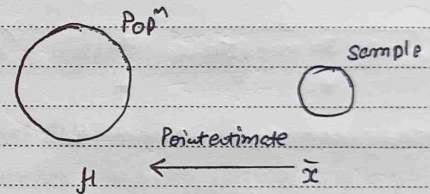
Estimates:

$\Rightarrow$ Estimate: It's specified observed numerical value used to estimate an unknown popⁿ parameter.

$\Rightarrow$ Types:

(a) Point estimate: single numerical value used to estimate an unknown popⁿ parameter.

eg, sample mean is a point estimate of popⁿ mean.



(b) Interval estimate: a range of values is used to estimate unknown popⁿ parameter.

